



Safe to swim?

Rapid Risk Forecasting for Horizons Regional Council Contact Recreation Sites

Ian Hurst – Team Leader Science | Environmental Monitoring

E. coli and Enterococci at the region's swimming spots

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We sample the water for the faecal indicator bacteria E. coli (in **rivers, streams, lakes, and estuaries**) or Enterococci (at **beaches**).

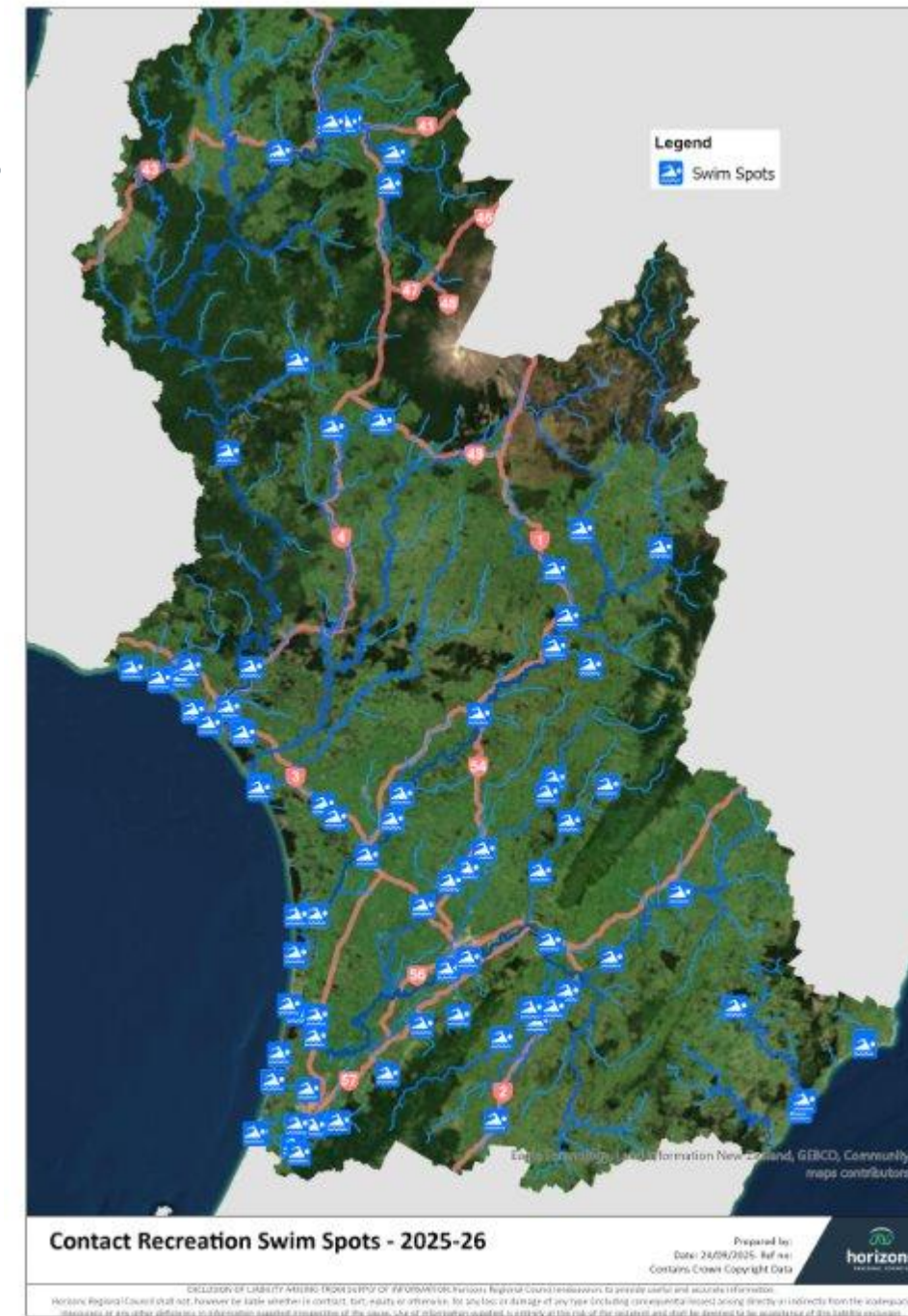


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These bacteria are commonly found in the gut of warmblooded animals and people. However, elevated levels of this bacteria indicate faecal (poo) contamination of the water, suggesting other disease-causing bacteria, viruses, and protozoa might also be present, increasing the risk that people playing in and on the water could become sick.



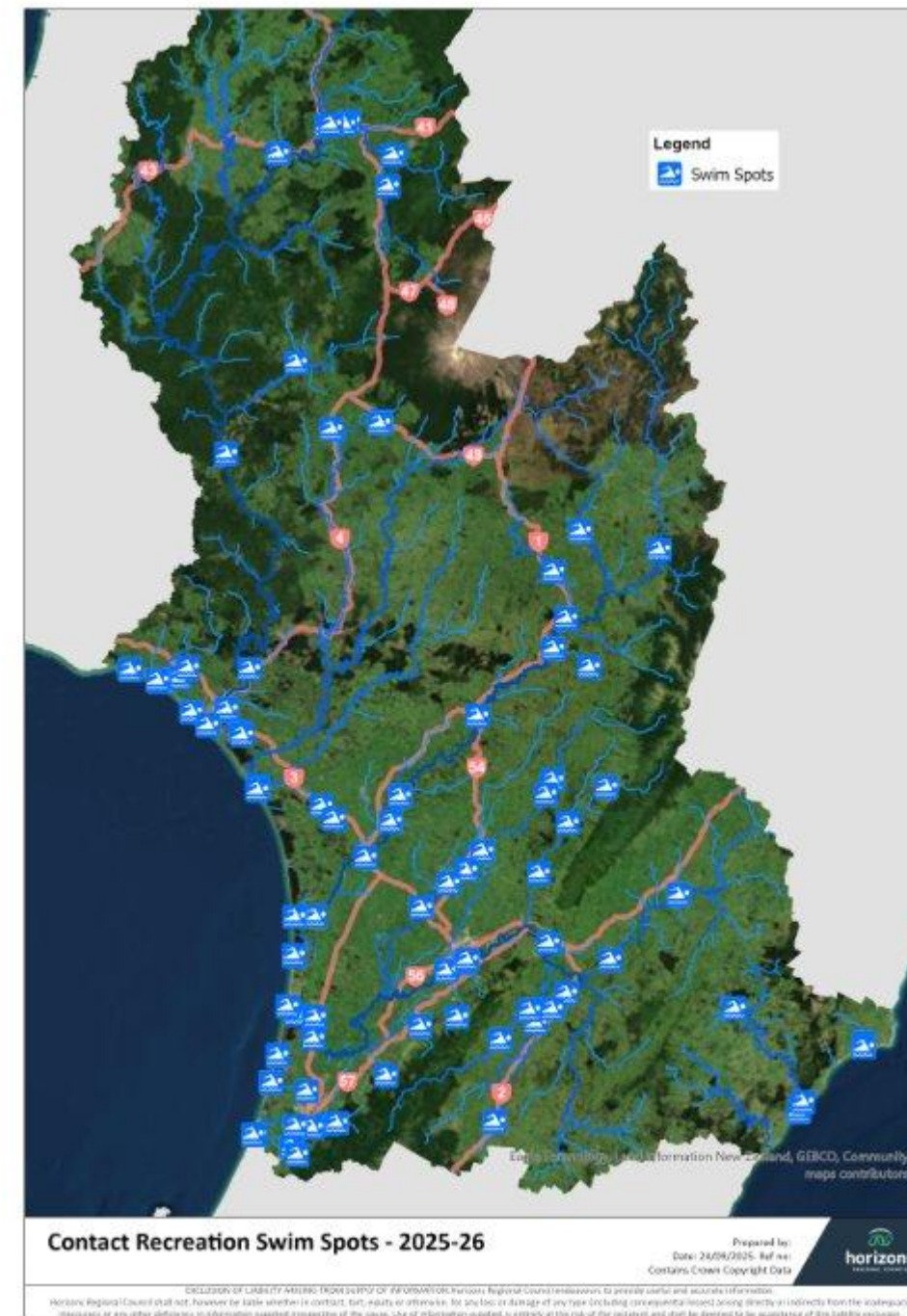
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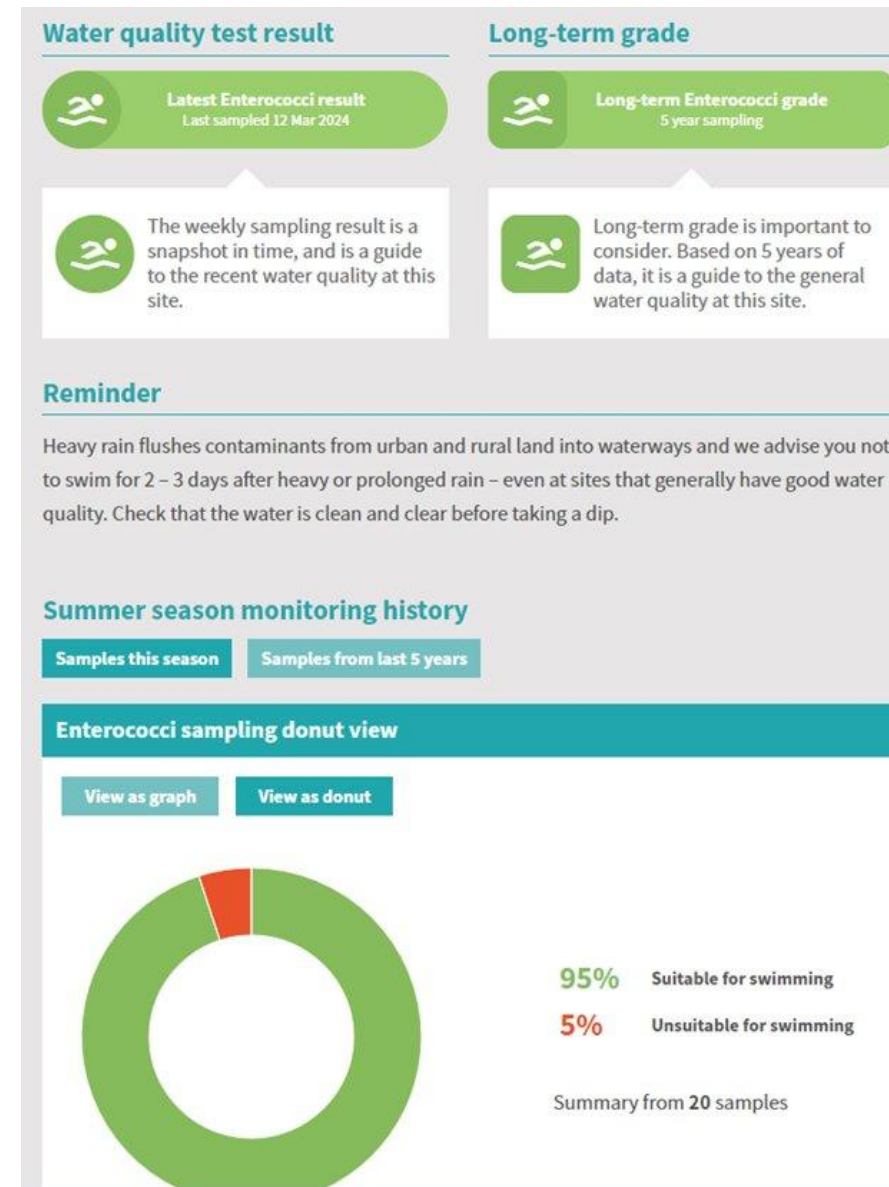
These bacteria enter waterways through animal and bird droppings, overland flow, and wastewater and stormwater discharges.



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www.LAWA.org.nz

The Land Air Water Aotearoa website (LAWA) publishes Regional and Unitary Councils' weekly recreational water quality “**swimmability**” results. Communities can check out their favourite swimming spot on LAWA to see if the results show the quality of the water is safe to play in.



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Suitable for swimming

Low risk of illness: This site met the water quality guidelines for faecal indicator bacteria and/or toxic algae when tested, OR the water quality is predicted to meet faecal indicator bacteria guidelines.



Caution advised

Elevated health risk: The faecal indicator bacteria were slightly elevated when tested, OR the predicted result indicates a slightly elevated health risk. While within the guidelines, young children, the elderly or individuals with health conditions may have an increased risk of illness.

For freshwater sites routinely monitored for toxic algae, this site exceeded the alert level criteria when tested. Avoid contact where toxic algae levels are elevated.



Unsuitable for swimming

High risk of illness: This site exceeded water quality guidelines for faecal indicator bacteria when tested, OR the water quality is predicted to exceed the guidelines.

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Mode	Trigger level		Management response
	Beach: Enterococci / 100mL		
 Surveillance	Equal to or less than 140 Enterococci / 100 mL		Routine monitoring.
 Alert	More than 140 Enterococci / 100 mL		Increase monitoring and investigate source.
 Action	More than 280 Enterococci / 100 mL		Public warnings if required, increased monitoring and investigation of contaminant source.



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Mode	Trigger level		Management response
	Beach: Enterococci / 100mL	River/Lake: <i>E. coli</i> /100 mL	
 Surveillance	Equal to or less than 140 Enterococci / 100 mL	Equal to or less than 260 <i>E. coli</i> / 100 mL	Routine monitoring.
 Alert	More than 140 Enterococci / 100 mL	More than 260 <i>E. coli</i> / 100 mL	Increase monitoring and investigate source.
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The issue:

Each site is monitored weekly

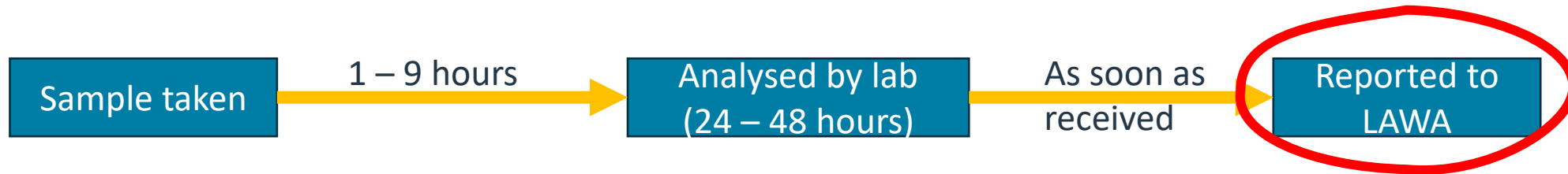


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Because water quality can change rapidly following heavy rain, this lag can result in the public being given out-of-date information that does not reflect current health risks.



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To bridge this gap, organisations are shifting toward models that synthesise historical sampling data with real-time environmental variables to predict swimmability on a more frequent basis.

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The Challenge:

Can we accurately classify the risk to swimmers in lieu of physical sampling, using environmental drivers such as rainfall, turbidity, and river flow?

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False Negatives: If the model predicts the water is safe when it is contaminated, there is a severe public health risk – Serious issue.



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4. Target update frequency is one prediction daily based on data from previous day(s).



Data and outcomes:

What we can provide:

- Weekly historical swim-spot data (Enterococci and *E. coli*) between Nov-Apr.
- Monthly year-around State of Environment data (*E. coli*, turbidity, water clarity).

NOT NECESSARILY SAME SITES

- River flow data, rainfall data.

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What we hope for from MINZ (some/all):

- Advice on large-scale (multi-site) or small-scale model (site-by-site) approach.
- Advice on what rainfall, flow sites best predict swimmability and at which monitoring sites.
- Initial model results and a clear implementation pathway.

The group will need to decide the scale of work this week (e.g. whole region, beach sites only, single catchment, etc).

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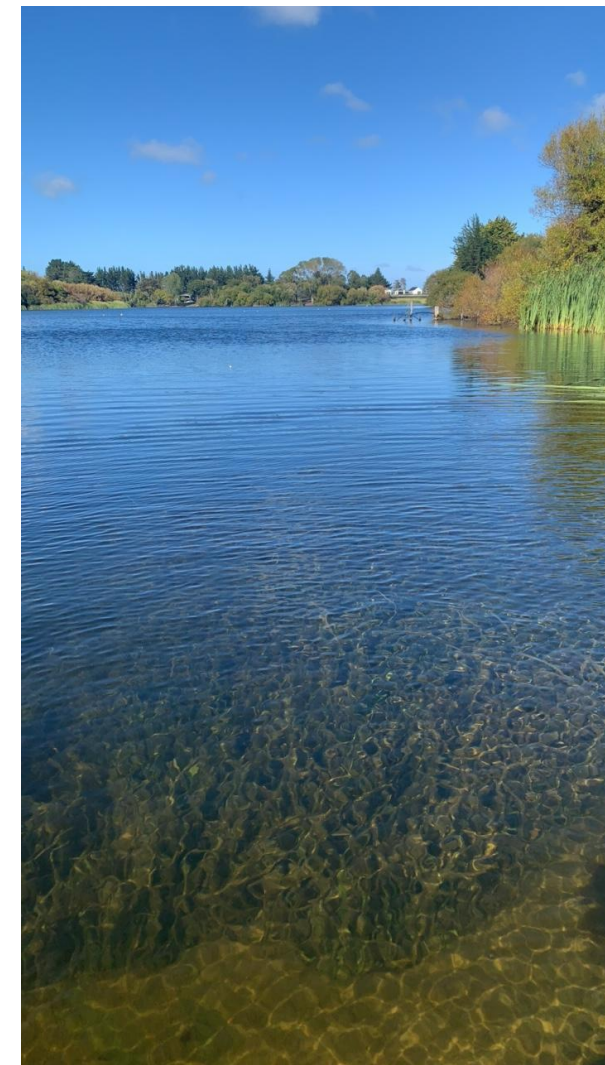
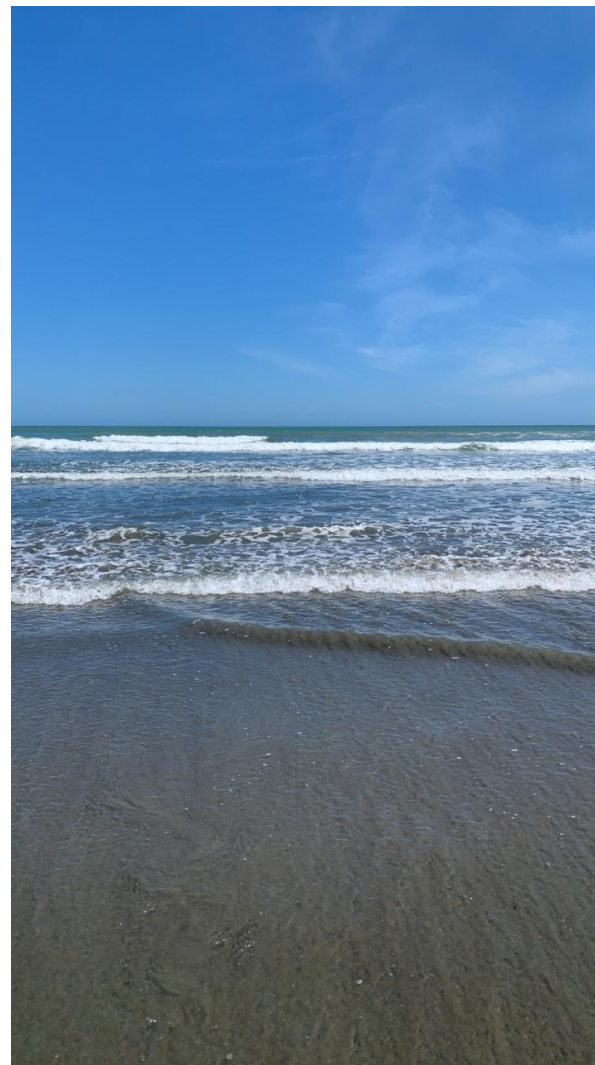
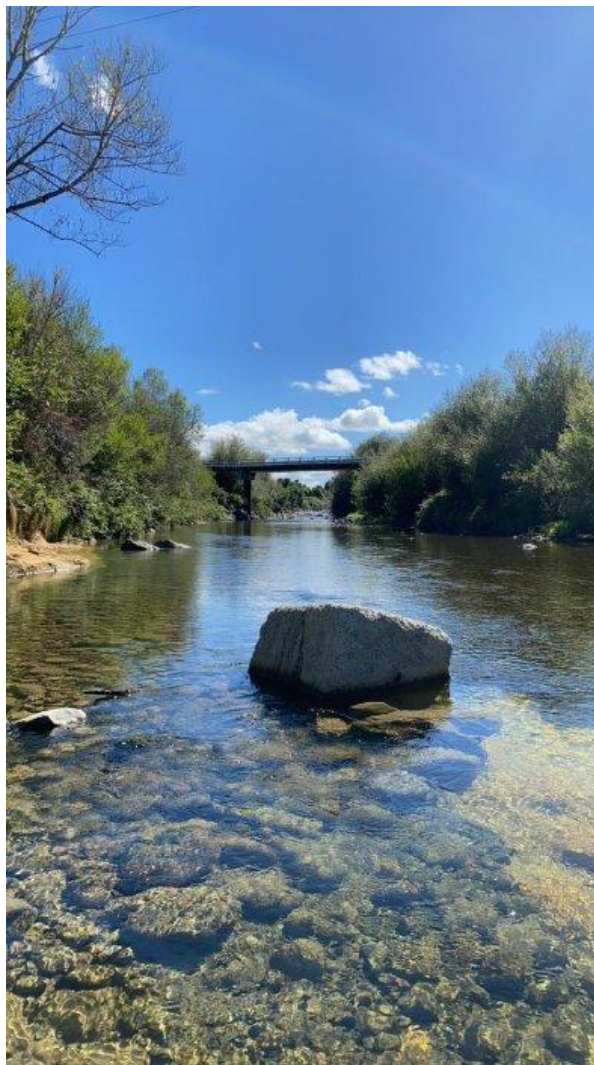
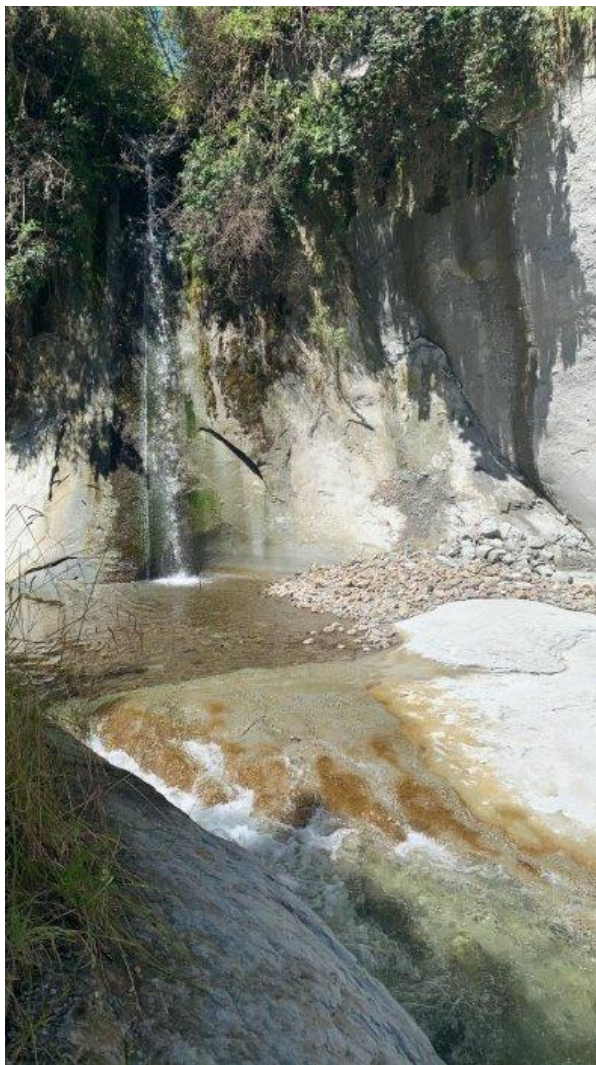
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Questions?





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REGIONAL COUNCIL



For more information visit www.horizons.govt.nz
or freephone Horizons on **0508 800 800**